

Lichun Is Not a Day: The Solar Term Is an Instant

Chujia Kuang (邝楚嘉)

嘉言一得 (Jiayan Yide)

Independent Researcher, Jieyang, Guangdong, China

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Chujia Kuang (邝楚嘉)

Independent Researcher, Jieyang, Guangdong, China

Corresponding author: Chujia Kuang, Independent Researcher, Jieyang, Guangdong, China. E-mail: chirca@163.com

Abstract

Lichun (立春), the beginning of spring, is ordinarily treated as a calendar day. It is not. It is an instant, defined in the national standard as the moment at which the sun's apparent geocentric ecliptic longitude reaches an integral multiple of 15 degrees. This note sets out what that definition entails in practice, and shows why the date of lichun nevertheless appears stable to within a day or two in ordinary years. The sixteen years from 2015 to 2030 are then examined: lichun falls on 3 February in four of them, in each case the first year after a leap year. The second half of the note traces the long preference for mean solar terms (平气, *pingqi*) over true solar terms (定气, *dingqi*) — proposed by Liu Zhuo in the sixth century, adopted for computation well before it was adopted for printed calendars, and not carried into the calendar itself until 1645. The note is descriptive; it contains no predictive claims.

Keywords: solar terms; lichun; ecliptic longitude; mean and true solar terms; calendar reform; Qing calendar

1. Twelve seconds, and half the notation changes

Take a concrete case.

In 2021, lichun fell at 22:58:39 on 3 February.

One second either side of that moment belongs to two different years and two different months in the stem-and-branch count:

- born at 22:58:38, one second before the term begins: 庚子 己丑 壬午 辛亥
- born at 22:58:50, eleven seconds after it begins: 辛丑 庚寅 壬午 辛亥

The interval between the two is twelve seconds. The day is the same, the double-hour is the same, and the later two designations are identical — but the year designation and the month designation have both changed.

This is not a contrived example. The same situation recurs every year around lichun, lixia, liqui and lidong, the four terms that begin the seasons.

2. A solar term is an instant

The national standard defines a solar term without ambiguity: it is the **instant** at which the sun's apparent geocentric ecliptic longitude equals an integral multiple of 15 degrees.

The operative word is **instant**.

Ecliptic longitude is the sun's position measured along the ecliptic. As set out in the preceding note, the sun completes one circuit in a year, and that circuit is divided into twenty-four terms of 15 degrees each. The substance of a solar term is therefore this: the sun has arrived at a particular position.

That moment has no relation whatever to which page the calendar has been turned to. A calendar is made for people to read; the sun does not consult it.

3. Why we remember it as a day

Because the Gregorian calendar and the solar terms both follow the sun. Two scales set against the same body will naturally keep step.

In the first half of the year the terms fall mostly around the 6th and the 21st of the month; in the second half, mostly around the 8th and the 23rd. The spread is a day or two at most.

Keeping step is not the same as being the same thing. A **stable date** is a consequence, not a cause. The cause is that the sun arrives at that position on time, and that the Gregorian year is itself laid out by the tropical year, so that the two happen to coincide.

Where they fail to coincide, the date moves.

4. Lichun falls on the 3rd in some years, the 4th in others

The moment of each term drifts later every year. The preceding note gave the figure: a few hours accumulate annually, and after some years the accumulation crosses midnight.

Over the sixteen years from 2015 to 2030, lichun falls on **3 February** in four of them: 2017, 2021, 2025 and 2029.

The interval is exactly four years, and in each case the year in question is the first after a leap year. February in a leap year carries one extra day, which shifts every subsequent term one step earlier in the calendar.

So the sentence "lichun is on 4 February" is wrong four times in sixteen years.

The 2021 case was especially close: the term fell at 22:58:39, leaving one hour and one minute before the day ended. A little later and it would have dropped to 4 February.

5. The long detour: mean terms and true terms

The earliest scheme divided the year into twenty-four equal intervals of time, each of 15.2 days. This was called the mean term (平气, pingqi). It was easy to compute and easy to lay out in a calendar.

The difficulty is that the sun does not move at a constant rate. The earth's orbit is an ellipse; it moves fastest in early January and slowest in early July, so the sun's apparent motion across the sky is correspondingly uneven. The equal-interval scheme assumes a constant rate, and therefore does not fit.

Liu Zhuo, in the sixth century, proposed the more accurate true term (定气, dingqi), fixed by the degree the sun has actually reached.

Yet when the Tang scholars compiled the *Dayan li*, they added a short note:

凡推日月行度及轨漏交蚀，并依定气。若注历即依恒气也。(In computing the courses of the sun and moon, and in the calculation of eclipses, the true terms are to be followed. In printing the calendar, the mean terms are still to be followed.)

Eclipses were computed by true terms; printed calendars continued to use mean terms.

To know which was correct and still use the incorrect one is explained by a single consideration: mean terms made the days easy to lay out. The compromise held until the end of the Ming. It was not reversed until the second year of Shunzhi in the Qing, 1645 — an interval of one thousand and forty-one years.

Once true terms were adopted, the intervals between terms ceased to be equal: around the winter solstice the gap is a little over fourteen days, and around the summer solstice it opens to nearly sixteen.

6. So the moment of a term must be recorded to the minute

A term is an instant, and anything determined by a term cannot be recorded to the day alone.

Where the moment falls within an hour or two of the term boundary, the minute is required as well.

This is why old calendars and present-day almanacs both report the moment of a term to the minute, and sometimes to the second.

One further point is easily confused. The moment of a term differs from year to year. When the data for the year in question are needed, the calendars compiled by the Purple Mountain Observatory are the domestic standard; impression is not a substitute. The same term may fall several hours apart in two consecutive years.

7. Closing note

A solar term is an instant, not a day.

As noted earlier, the popular reckoning of the zodiac sign by the first day of the first month and the reckoning of the year by lichun run on two different conventions. This note supplies the second half of that statement: lichun is itself only an instant, and a few minutes either way can mean the difference between two designations of the year.

So when a date is reported without a time, the necessary question is: at what hour and minute?

What follows from this is the double-hour — and the double-hour does not begin at midnight. It is two hours long, and it crosses into the following day.

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Language. The original Chinese text is deposited as a companion file to this record (立春不是一整天，只有一秒钟).

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